

Phase 6 — Analytical Model Validation Report

Project: Retail Sales Performance Analytics

Database: retail_sales_analytics

Database Engine: PostgreSQL

Validation Script: sql/data_modeling_validation.sql

1. Executive Validation Result

PASS — ALL PHASE 6 VALIDATION CHECKS PASSED

A total of 40 analytical-model validation checks were executed. All 40 checks passed and 0 checks failed.

Validation Area	Checks	Passed	Failed
Row Count	8	8	0
Fact Grain	4	4	0
Primary Key	4	4	0
Foreign Key	8	8	0
Reconciliation	7	7	0
Business Totals	3	3	0
Key Completeness	4	4	0
Date Dimension	2	2	0
TOTAL	40	40	0

2. Row-Count Validation

Source → Analytical Table	Source	Target	Difference	Status
Customers → dim_customer	99,441	99,441	0	PASS
Sellers → dim_seller	3,095	3,095	0	PASS
Products → dim_product	32,951	32,951	0	PASS
Orders → fact_orders	99,441	99,441	0	PASS
Order Items → fact_order_items	112,650	112,650	0	PASS
Payments → fact_payments	103,886	103,886	0	PASS
Reviews → fact_reviews	99,224	99,224	0	PASS
Geolocation → dim_geography	19,015	19,015	0	PASS

Geography is validated at the analytical distinct ZIP-prefix grain rather than raw geolocation row count, consistent with the modeled deduplicated geography dimension.

3. Grain and Key Integrity

Fact-grain validation returned zero duplicate records for: fact_orders (order_id), fact_order_items (order_id + order_item_id), fact_payments (order_id + payment_sequential), and fact_reviews (review_key).

Primary-key validation returned zero duplicate keys for dim_customer, dim_seller, dim_product, and fact_orders.

4. Foreign-Key / Orphan Validation

All eight tested analytical relationships returned zero orphan records. Validated relationships include fact_orders to dim_customer and dim_order_status; fact_order_items to dim_product, dim_seller, and dim_date; fact_payments to dim_customer; and fact_reviews to dim_customer and dim_date.

5. Phase 5 Source-to-Analytics Reconciliation

Business Key / Grain	Source	Analytics	Difference	Status
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Customer IDs	99,441	99,441	0	PASS
Product IDs	32,951	32,951	0	PASS
Seller IDs	3,095	3,095	0	PASS
Order IDs	99,441	99,441	0	PASS
Order Items	112,650	112,650	0	PASS
Payments	103,886	103,886	0	PASS
Reviews	99,224	99,224	0	PASS

6. Business-Total Reconciliation

Metric	Source Total	Analytics Total	Difference	Status
Payment Value	16,008,872.12	16,008,872.12	0.00	PASS
Order Item Price	13,591,643.70	13,591,643.70	0.00	PASS
Freight Value	2,251,909.54	2,251,909.54	0.00	PASS

7. Key Completeness and Date Dimension

All four mandatory fact-key completeness checks passed with zero NULL values: fact_orders.customer_key, fact_orders.order_status_key, fact_order_items.product_key, and fact_order_items.seller_key.

The date dimension passed both uniqueness checks: no duplicate date_key values and no duplicate full_date values.

8. Final Conclusion

The Phase 6 analytical/star-schema model has been validated against the populated PostgreSQL analytical layer and the Phase 5 source data. The executed validation evidence shows 40/40 checks passed with zero failures. The model preserves required row counts, expected fact grains, key integrity, source business-key coverage, and core financial totals. No validation exception remains from the executed Phase 6 validation run.

Evidence source: Console execution of sql/data_modeling_validation.sql. The report records the validation results supplied from that execution.